

APPLICATIONS

EVALUATING BATSMEN'S SCORES IN TEST CRICKET

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Summary: Run accumulation by a batsman in test or first-class cricket is modelled by means of a Simple Geometric Distribution, and batting performance examined in (batting average, batting consistency) space. A statistic for adjusting the batting average for consistency, or what we term the Geometric Coefficient in this context, is proposed as an appropriate adjustment factor to incorporate a measure of a batsman's consistency into his batting average. Two sets of data, one pre-WWII and one current, are examined in this framework.

1. Introduction

In the game of cricket a batsman's *batting average*, which is a measure of the average number of runs scored per completed innings, has traditionally been the only calculated and documented statistic available to indicate his ability as a batsman. In recent years it has been recognised that in one-day limited-overs cricket a batsman's *strike rate* (i.e. the number of runs scored per ball faced) is probably a more important indicator, since in that version of the game the rate at which runs are accumulated by a batsman has become more important than the number of runs scored. This situation, in which the number of balls remaining

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in a side's innings is limited to n , has been the subject of a recent paper by Barr, Zucchini and Kantor (1997), who model how strategy might be optimised under the ODI format. The paper establishes how under certain assumptions run accumulation is expected to follow a Truncated Geometric Distribution.

However, test cricket, in which a game spans over five days (or alternatively a first-class match played over four days), is still considered to be the "real" game, and it is on this form of the game that this paper concentrates. In this case the match is effectively "open-ended", and a batsman may bat until he is out. Run accumulation is then expected to follow a Simple Geometric Distribution. Early work by Wood (1945) has shown that the Geometric Distribution can be used to model run accumulation quite accurately.

A batsman's batting average uses the data available in a one-dimensional manner. This definition of a batsman's prowess, whilst adequately comparing batsmen in terms of their ability to score runs, is lacking in that it does not reflect the batsman's *consistency* of scoring runs. This extra information may be useful to the coach of a test team, who might prefer to select a batsman who scored, say, 40 runs every innings to a batsman whose score was uniformly distributed on the interval $[0;80]$. The feature of exceptional levels of consistency can be indicated by a batsman "breaking away" from the Geometric Distribution. A plot of mean (i.e. batting average) against standard deviation of scores for an exceptionally consistent batsman will be indicated as a mean *larger* than that expected (under the Geometric) for the given standard deviation.

In this paper we attempt to model the accumulation of runs by a batsman; we develop, under some mild assumptions, the statistical distribution of the number of runs scored, and we use the moments of this distribution to arrive at a statistic which appears to be a suitable adjustment measure to incorporate

consistency into the calculated batting average. In the next section we develop a model of run accumulation in open-ended cricket. In Section 3 this model is tested on a pre-World War II data set, as well as the test batting averages of players in the current (1997/98) South African national squad.

2. Modelling run accumulation in open-ended cricket

In order to model the process of run accumulation by a batsman we consider a scenario with a set of well defined conditions. Quite clearly, these conditions will not be completely fulfilled, but they may well be fulfilled to the extent that the modelling process gives very real insight into what parameters are critical in the process of run accumulation. We assume, in the first instance, that each ball faced by a batsman can be regarded as a Bernoulli experiment, in which the outcomes are either "gets out" or "does not get out". We also assume that the trials (i.e. the balls received by the batsman) are independent and that the probability of getting out, p , is constant. In addition, we assume that the batsmen bat "one at a time". That is, Batsman 1 goes in and faces all the balls until he gets out, then Batsman 2 goes in and faces all the balls until he gets out, and so on. Clearly these assumptions are simplifications of reality: for example, a bowler may change the type and distribution of balls delivered according to the array of shots played by a particular batsman and hence p may vary from ball to ball. We will assume, though, that keeping p constant in the model is a good approximation of reality.

In order to develop a model of run accumulation we introduce the following notation:

- R number of runs scored by a given batsman in a given innings
 X the number of balls faced (excluding one that (might) get the batsman out)
 S strike rate of the batsman (the number of runs scored per ball faced (excluding the one that (might) get the batsman out))
 p probability of the batsman getting out
 $q = 1 - p$ = probability of not getting out.

In the open-ended cricket case, and under the assumptions above, for any batsman the random variable X will have a Geometric Distribution with parameter p ; i.e. $X \sim GD(p)$. Thus X has probability function given by

$$p(x) = pq^x \quad x = 0, 1, 2, \dots$$

By definition, the expectation of the number of balls faced, X , is given by

$$E(X) = \frac{q}{p}$$

Furthermore, the standard deviation of X is given by

$$\sigma(X) = \frac{\sqrt{q}}{p},$$

implying that

$$CV(X) = \frac{1}{\sqrt{q}}.$$

From the definitions given above, we have that $R = S * X$. Therefore the expected number of runs scored by a batsman is given by

$$E(R) = S * E(X)$$

$$\begin{aligned}
 &= \frac{\text{Total number of runs scored}}{\text{Total number of (not out) balls faced}} \cdot \frac{\text{Total number of (not out) balls faced}}{\text{Total number of completed innings}} \\
 &= \text{Batting average}
 \end{aligned}$$

Note that this expectation is independent of the number of completed trials (i.e. balls received). Thus, at any point in a batsman's innings his expectation of the score he will achieve is his average score. That is, the distribution of runs yet to be scored is independent of the state of his innings; the distribution thus exhibits the so-called "loss of memory" property.¹

Furthermore

$$\sigma(R) = S * \frac{\sqrt{q}}{p}$$

and thus

$$CV(R) = \frac{1}{\sqrt{q}}.$$

A crude estimate of q (the probability of not being out on a particular ball) is thus²

$$\hat{q} = (\widehat{CV(R)})^{-2}.$$

Thus, if a batsman's scores follow a Geometric Distribution, the inverse of the square of the CV of the number of runs scored, which measures the number of standard deviations by which the mean differs from zero, is thus a measure of the batsman's probability of not going out on a particular ball; the higher the value of q the less the chance of a batsman going out on any particular ball. Note that the initial assumptions led to the random variable X (the number of balls faced) having a Geometric Distribution, and this implies that q is less than unity. If a batsman is capable of being unusually consistent then the assumptions

¹ Wood (1945) examined a number of batsmen over the period 1890–1939 including all the "great" (op. cit. p. 14) batsmen of the period. He came to the conclusion that the geometric distribution fairly approximated the distribution of their scores. The paper was the focus of much discussion in the Royal Society meeting of November 28th 1944, including a contribution by the famous English captain Sir P.F. Warner.

² A problem with the past recording of a batsman's performance in test cricket is that the number of balls faced has only rarely been recorded. (In the case of ODI cricket, where strike rate is so important, the number of balls faced has usually been recorded.) If the batsman's scores can be assumed to follow a Geometric Distribution this is thus one way of estimating q in first class or test (open-ended) cricket where the number of balls faced is invariably unknown.

of the Geometric will not apply. In this case the expected number of runs may exceed the standard deviation of runs scored by a significant margin.

We have found it useful when looking at batting performance to consider scores in so-called risk-return space, an approach commonly used in Financial Analysis. Then if we plot $E(R)$ on the vertical axis and $\sigma(R)$ on the horizontal axis, then the ratio $\frac{E(R)}{\sigma(R)}$ is a measure of risk-adjusted return, or batting consistency.³ Figure 1 indicates a plot in $(E(R), \sigma(R))$ space of what one might expect hypothetically. If batsmen are represented by crosses and they conform quite closely to a Geometric Distribution one would expect them to generally lie close to the "GD curve" (Geometric Distribution curve). Since this curve represents points equal to $\sqrt{q}$ for a range of $\frac{E(R)}{\sigma(R)}$, and as $\sqrt{q}$ will tend to 1 for good batsmen (where the probability of getting out is low), this curve may be expected to curve upwards and approach the 45 degree line (the line where $\frac{E(R)}{\sigma(R)} = 1$) as one moves from left to right.

³ This ratio is often used in Financial Analysis and termed the Sharp Ratio (Sharp, 1970).

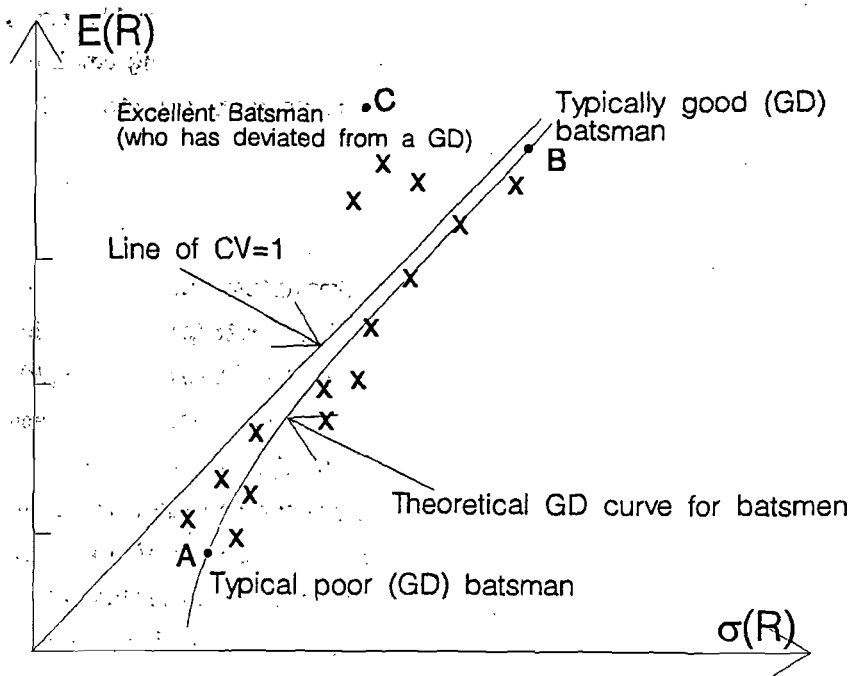


Figure 1. Hypothetical representation in $(E(R), \sigma(R))$ space of batting ability

Points plotted in $(E(R), \sigma(R))$ space which lie *above* the 45 degree line require a rather different characterisation. The measure $\frac{E(R)}{\sigma(R)}$ is clearly a measure of consistency of scoring – we call it the Geometric Coefficient in this context. As one moves vertically along an iso-sigma line, a batsman's average increases, as does his consistency (relative to this average). Above the 45 degree line, however, one is necessarily deviating from the Geometric

Distribution. Batsmen who have averages above the 45 degree line can thus be interpreted as having averages which are over and above those which might be expected for their level of riskiness. That is, they are *unusually consistent* for their average and in fact have levels of consistency which are *not consistent* with a Geometric Distribution.

3. Illustrative Data

Table 1 contains estimates of $E(R)$, (we call these estimates $Average(R)$) and estimates of $\sigma(R)$, (we call these estimates $S(R)$) for a set of pre-WWII batting data collected by Woods (1945). This data is plotted in $(Average(R), S(R))$ space in Figure 2 (complete data) and Figure 3 (excluding Bradman and Barlow). Each batsman had scored in excess of ten thousand runs in first class cricket. It is seen how closely the set of batsmen track the 45 degree line. The two who lie above the line are in fact the two super batsmen of that era, D. Bradman and J.B. Hobbs. They have broken free of the Geometric in their measure of consistency (relative to their batting average). The characterisation of Bradman's supreme talent is thus emphasised. Not only was he scoring at an unprecedented average, but his consistency was higher than *that which could be expected for that average*.

The statistic $\frac{Average(R)}{S(R)} = \frac{Batting\ Average}{Standard\ deviation\ of\ scores}$ or Geometric Coefficient (GC), effectively measures the extent to which a batsman has broken free of the Geometric, and may thus prove a useful factor for capturing a batsman's ability to raise his level of consistency over *that which may have been expected*.⁴ Thus weighting the batting average by the Geometric

⁴ Note that if the batsman's scores follow a geometric distribution, then GC^2 could be used to estimate q . However, we are primarily trying to estimate the deviation of a batsman's scores from the Geometric rather than q itself.

Coefficient value is a way of incorporating consistency into the reported average. In Table 1 the column headed "Adjusted Average" contains the consistency-adjusted averages for Woods' set of batsmen. Note that in all cases except Hobbs and Bradman the GC is less than unity, implying that the consistency-adjusted batting average is *less* than the conventional average. Table 2 contains the average test scores and standard deviations for all the players in the SA national squad as at the end of November 1997. These are depicted in $(Average(R), S(R))$ space in Figure 4. This data, based on relatively few innings by each player, shows that in 12 cases $GC > 1$ (indicating that these players were particularly consistent) and in only 6 cases was $GC < 1$. Having a high proportion of very consistent batsmen is a likely factor in ensuring a successful test team; this fact sets this data set apart from Woods' data. Some individual batsmen deserve particular discussion. Pollock has an exceptionally high GC of 1.48, which raises his average from 35.71 (a good average for a lower middle-order batsman) to 52.87. Although still relatively young and inexperienced in test cricket, many authorities of the game have singled Pollock out as a player with exceptional talent. Cullinan (average = 37.71) and McMillan (average = 39.52) have similar batting averages, but their GCs differ markedly (Cullinan: $GC = 0.79$; McMillan: $GC = 1.21$), leaving their consistency-adjusted averages at 29.67 and 47.80 respectively. At approximately this point in time Cullinan was dropped from the team, an indicator that his inconsistency had been noted by the selectors. In contrast McMillan, whose recent scores had been exceptionally disappointing, was retained. Comparing Figure 4 to the hypothetical representation in Figure 1, it can be seen that other "good" batsmen are Kirsten, Cronje and Bacher, whilst Kallis, De Villiers and Adams have performed poorly as batsmen (Adams and

De Villiers were in the team primarily as bowlers, so were not expected to excel as batsmen).

4. Conclusion

The use of a batting average in assessing an individual batsman's performance is flawed from a number of perspectives, most particularly because it takes no account of batting consistency. In this paper we propose and implement for two data sets a measure that adjusts the batting average by a measure of consistency. The actual adjustments are in line with intuition regarding particular batsmen. The rationale for the adjustment procedure stems from the assumption that batting scores in test and first-class cricket follow a Geometric Distribution.

5. References

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Table 1. Average scores, estimated standard deviation, estimated CV, GC and adjusted averages for a group of pre-WWII batsmen (from Woods (1945)) in first class matches

	Average(R)	S(R)	CV(R)	GC*100	Adjusted Average
R.G. Barlow	18.80	19.70	104.7	95.51	17.96
K.S. Ranjitsinhji	49.50	52.70	106.4	93.98	46.52
C.B. Fry	46.70	49.20	105.4	94.88	44.31
J.B. Hobbs	45.70	44.10	96.2	103.95	47.51
W.R. Hammond	50.10	56.60	113	88.50	44.34
H. Sutcliffe	46.30	47.60	102.8	97.28	45.04
C.P. Mead	40.80	43.70	107.1	93.37	38.10
E.E. Woolley	38.40	41.20	107.3	93.2	35.79
E. Hendren	44.60	47.80	107.4	93.11	41.53
J.W. Hearne	36.70	42.40	116.1	86.13	31.61
W.G. Grace	36.80	42.80	116.4	85.91	31.62
T. Hayward	37.90	41.50	109.5	91.32	34.61
P.F. Warner	33.70	36.30	107.6	92.94	31.32
C.J. Burnup	34.90	35.60	101.9	98.14	34.25
F.S. Jackson	31.60	33.40	105.7	94.61	29.90
M. Leyland	36.90	40.30	109.2	91.58	33.79
J.T. Tyldesley	38.40	41.90	109.1	91.66	35.20
L.E.G. Ames	38.40	41.90	109.1	91.66	35.20
G. Gunn	33.20	34.10	102.7	97.37	32.33
E. Tyldesley	40.20	42.60	106	94.34	37.92
C.B. Harris	31.40	33.00	105.1	95.15	29.88
D. Bradman	84.00	83.20	99.1	100.91	84.76

Table 2. Average scores, estimated standard deviation, estimated CV, GC and adjusted averages for all test innings for current SA national squad players as at November 1997

	Average(R)	S(R)	CV(R)	GC*100	Adjusted Average
H. Cronjé	35.27	32.34	0.92	109.06	38.46
A. Bacher	33.07	32.03	0.97	103.25	34.15
H. Gibbs	15.63	12.18	0.78	128.32	20.05
A. Hudson	34.91	38.12	1.09	91.58	31.97
D. Cullinan	37.71	47.93	1.27	78.67	29.67
G. Kirsten	38.22	33.81	0.88	113.06	43.22
J. Kallis	14.75	22.66	1.54	65.08	9.60
B. McMillan	39.52	32.68	0.83	120.95	47.80
J. Rhodes	29.52	26.88	0.91	109.82	32.42
D. Richardson	25.94	25.58	0.99	101.42	26.31
P. Adams	5.38	29.99	5.57	17.96	0.97
F. de Villiers	17.94	26.63	1.48	67.37	12.09
A. Donald	12.29	8.80	0.72	139.57	17.15
L. Klusener	33.00	27.32	0.83	120.81	39.87
C. Mathews	18.32	15.64	0.85	117.12	21.45
S. Pollock	35.71	24.12	0.68	148.04	52.87
B. Schultz	1.50	2.10	1.40	71.42	1.07
P. Symcox	26.06	23.09	0.89	112.88	29.42

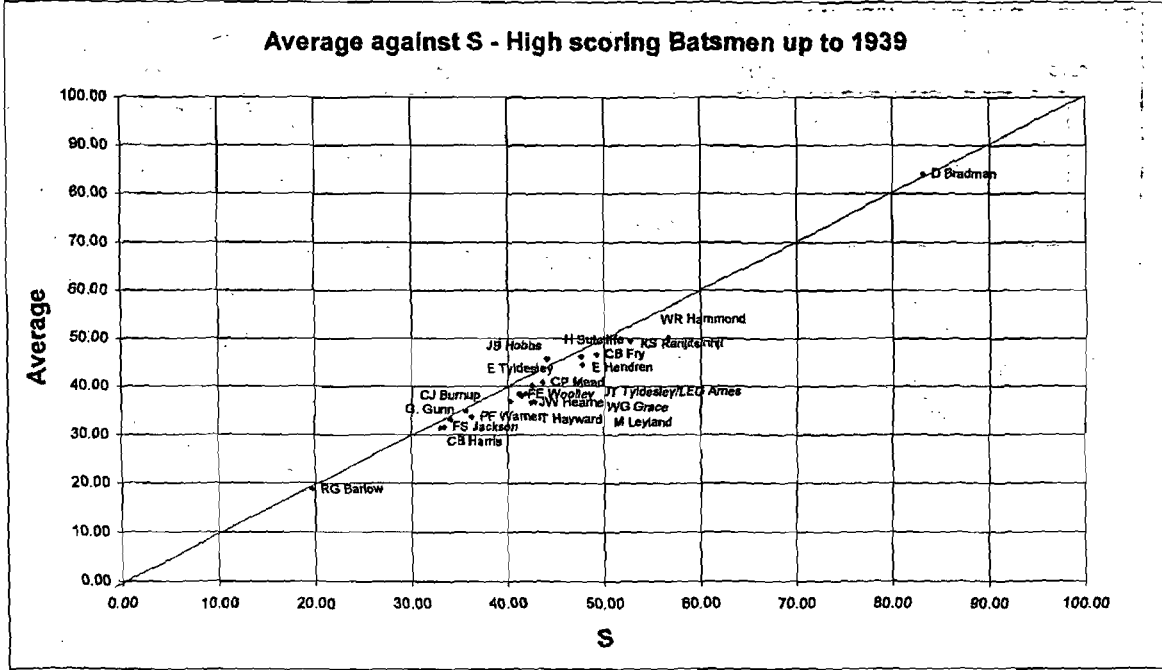
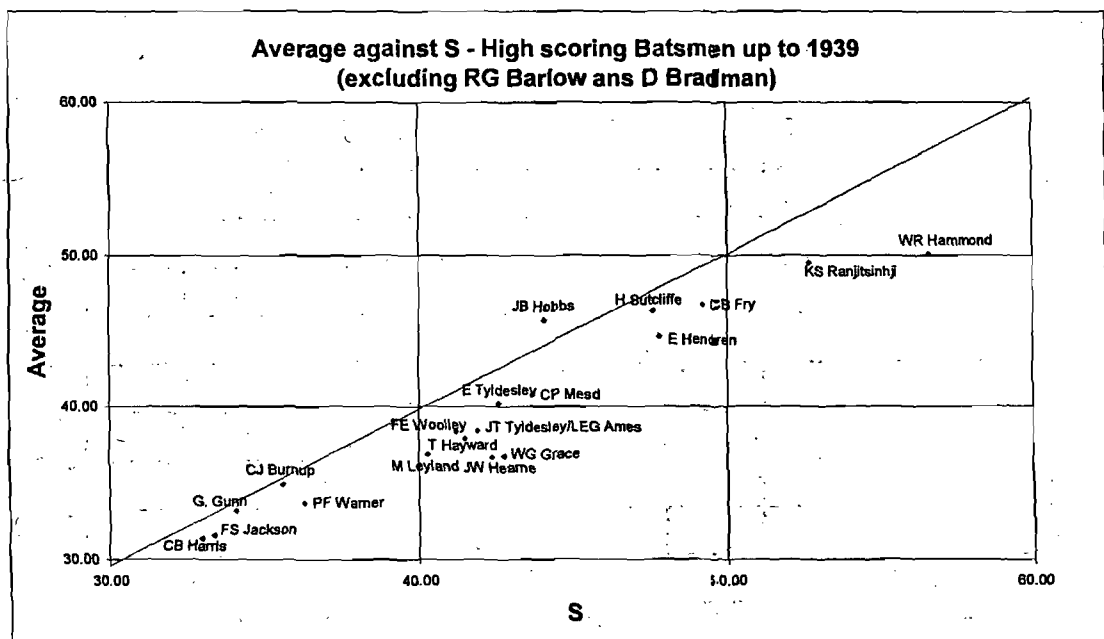


Figure 2

Figure 3



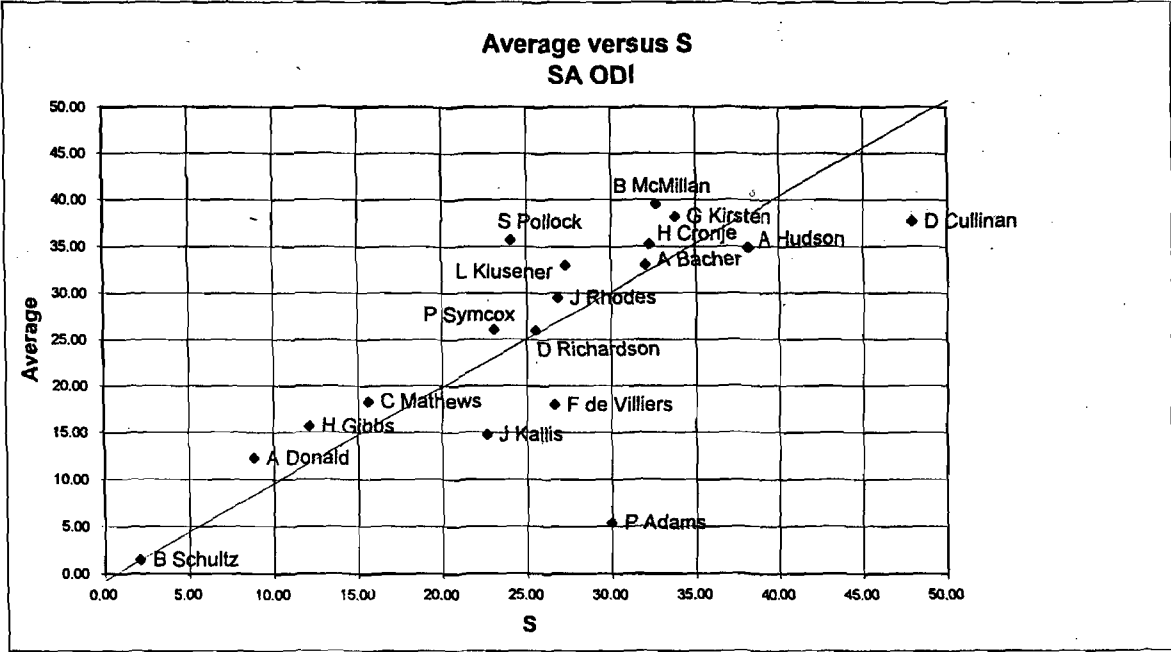


Figure 4